Solving My Open Problem

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Abstract

In [1], I posed the following open problem: given the sequence

$$s_n = \int_{[0,1]^n} \prod_{i=1}^{n} x_i^{x_i} \, dx_1 \cdots dx_n,$$

does $\lim_{n\to\infty} s_n$ exist, and if so, what is its value? We answer both questions definitively. Using Fubini's theorem and an elementary induction we show $s_n = c^n$, where $c = \int_0^1 x^x \, dx \approx 0.7834$. Since $c \in (0,1)$, a classical ε-N argument then gives $\lim_{n\to\infty} s_n = 0$. We further generalise the problem to arbitrary permutations $\sigma \in S_n$ of the exponents, prove that the identity permutation is the unique maximiser among all σ via the Rearrangement Inequality, and establish that the limit equals zero in every permutation case. Algorithmic aspects, numerical evidence, and a cycle-decomposition analysis complete the picture.

The paper ends with "The End"

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1 Introduction

Open problems that admit surprisingly clean solutions often illuminate deep connections between otherwise distant branches of mathematics. My problem [1] is one such instance: what appears at first glance to be a complex limit of nested integrals resolves, once the right perspective is adopted, into an elegant interplay of measure theory, real analysis, combinatorics, and inequalities.

The integrand x^x is itself a classical object. It appears in the well-known "sophomore's dream" identities [2]:

$$\int_0^1 x^{-x}\,dx = \sum_{n=1}^{\infty} n^{-n}, \tag{1}$$

$$\int_0^1 x^{x}\,dx = -\sum_{n=1}^{\infty} (-n)^{-n}. \tag{2}$$

The constant $c = \int_0^1 x^x\,dx$ therefore has a well-known series representation, and its numerical value is known to high precision [5].

The present paper is organised as follows. Section 2 recalls the necessary background in measure theory and combinatorics. Section 3 proves the separation of the n-fold integral by induction, using Fubini's theorem rigorously. Section 4 gives the ε-N proof of the limit. Section 5 analyses the general permutation case and establishes the Rearrangement-Inequality-based maximality of the identity permutation. Section 6 provides numerical evidence and algorithm descriptions. Figures and tables corroborate the theoretical results throughout.

2 Preliminaries

2.1 Notation

Throughout, $[0,1]^n$ is equipped with the n-dimensional Lebesgue measure λ^n. The symmetric group on n letters is S_n; its elements are written $\sigma, \tau, \ldots$. The cycle type of σ is the partition $(k_1, \ldots, k_m)$ with $k_1 + \cdots + k_m = n$. We write $|\sigma|$ for the number of cycles of σ.

2.2 Key Analytical Tools

Theorem 2.1 (Fubini–Tonelli)**.** *Let $(X, \mathcal{A}, \mu)$ and $(Y, \mathcal{B}, \nu)$ be σ-finite measure spaces. If $f : X \times Y \to \mathbb{R}$ is $(\mathcal{A} \otimes \mathcal{B})$- measurable and either non-negative or $\mu \otimes \nu$-integrable, then*

$$\int_{X\times Y} f\,d(\mu\otimes\nu) \;=\; \int_X \left(\int_Y f(x,y)\,d\nu(y)\right) d\mu(x) \;=\; \int_Y \left(\int_X f(x,y)\,d\mu(x)\right) d\nu(y).$$

Theorem 2.2 (Rearrangement Inequality [3])**.** *Let $a_1 \le a_2 \le \cdots \le a_n$ and $b_1 \le b_2 \le \cdots \le b_n$ be real sequences. For any permutation $\sigma \in S_n$,*

$$\sum_{i=1}^{n} a_i b_i \;\ge\; \sum_{i=1}^{n} a_i b_{\sigma(i)},$$

with equality if and only if σ preserves the order of the b_i on the support where all a_i are distinct.

2.3 The Constant c

Definition 2.3. *We write*

$$c \;:=\; \int_0^1 x^x \, dx.$$

Here $x^x := e^{x \ln x}$ for $x \in (0,1]$, and the integrand is extended by continuity to $x = 0$ via $\lim_{x\to 0^+} x^x = 1$.

Lemma 2.4. $c \in (0,1)$.

Proof. Positivity. The function $x \mapsto x^x$ is continuous and strictly positive on $(0,1]$, with value 1 at $x = 0$ by the extension above. Hence $c > 0$.

Strictly less than one. For $x \in (0,1)$ one has $\ln x < 0$, so $x \ln x < 0$ (since $x > 0$), hence $x^x = e^{x \ln x} < e^0 = 1$. Therefore $c = \int_0^1 x^x \, dx < \int_0^1 1 \, dx = 1$. □

The numerical value, computed via the series (2) or by numerical quadrature (see Section 6), is

$$c \approx 0.78343\,05107\,12\ldots$$

3 Separation of the n-Fold Integral

Theorem 3.1 (Product Separation)**.** *For all $n \ge 1$,*

$$s_n \;=\; \int_{[0,1]^n} \prod_{i=1}^{n} x_i^{x_i} \, dx_1 \cdots dx_n \;=\; c^n.$$

Proof. We proceed by induction on n.

Integrability prerequisite. Since $x^x \in [(1/e)^{1/e}, 1]$ for all $x \in [0,1]$, the integrand satisfies $0 < \prod_{i=1}^{n} x_i^{x_i} \le 1$ everywhere on $[0,1]^n$. Being a finite product of Borel-measurable functions, the integrand is itself Borel-measurable and bounded, hence Lebesgue integrable. Fubini's theorem (Theorem 2.1) therefore applies at every step below.

Base case ($n = 1$). $s_1 = \int_0^1 x^x \, dx = c = c^1$ by Definition 2.3. ✓

Inductive step. Suppose $s_n = c^n$ for some $n \ge 1$. Then

$$\begin{aligned}
s_{n+1} &= \int_{[0,1]^{n+1}} \prod_{i=1}^{n+1} x_i^{x_i} \, dx_1 \cdots dx_{n+1} \\
&= \int_0^1 \left(\int_{[0,1]^n} \prod_{i=1}^{n+1} x_i^{x_i} \, dx_1 \cdots dx_n \right) dx_{n+1} && \text{(Fubini, Theorem 2.1)} \\
&= \int_0^1 x_{n+1}^{x_{n+1}} \left(\int_{[0,1]^n} \prod_{i=1}^{n} x_i^{x_i} \, dx_1 \cdots dx_n \right) dx_{n+1} && (x_{n+1}^{x_{n+1}} \text{ constant w.r.t. } x_1, \ldots, x_n) \\
&= \int_0^1 x_{n+1}^{x_{n+1}} \cdot s_n \, dx_{n+1} && \text{(definition of } s_n) \\
&= c^n \int_0^1 x_{n+1}^{x_{n+1}} \, dx_{n+1} && \text{(inductive hypothesis; } c^n \text{ is constant)} \\
&= c^n \cdot c \;=\; c^{n+1}.
\end{aligned}$$

□

4 The Limit: an ε-N Proof

Theorem 4.1 (Main Result)**.** $\lim_{n\to\infty} s_n = 0$.

Proof. By Theorem 3.1, $s_n = c^n$ where $c \in (0,1)$ by Lemma 2.4. Thus $\ln c < 0$.

Let $\varepsilon > 0$ be arbitrary. Choose

$$N \;:=\; \left\lfloor \frac{\ln \varepsilon}{\ln c} \right\rfloor + 1 \;\in\; \mathbb{N}.$$

Then for any $n > N$:

$$\begin{aligned} & n > \frac{\ln \varepsilon}{\ln c} \\ \implies \quad & n \ln c < \ln \varepsilon \qquad \text{(multiply by } \ln c < 0\text{; inequality reverses)} \\ \implies \quad & c^n < \varepsilon \\ \implies \quad & |s_n - 0| = c^n < \varepsilon. \end{aligned}$$

Since $\varepsilon > 0$ was arbitrary, $s_n \to 0$ as $n \to \infty$. □

Remark 4.2. *The ε-δ formulation is reserved for limits over continuous domains. Because n ranges over $\mathbb{N}$, the ε-N formulation is the correct rigorous framework, and is entirely analogous in structure.*

5 The General Permutation Case

5.1 Setup

Definition 5.1. *For $\sigma \in S_n$, define*

$$s_n^\sigma \;:=\; \int_{[0,1]^n} \prod_{i=1}^{n} x_i^{x_{\sigma(i)}}\, dx_1 \cdots dx_n.$$

The original sequence satisfies $s_n^{\mathrm{id}} = s_n = c^n$.

5.2 Cycle Decomposition and Factorisation

Every permutation $\sigma \in S_n$ decomposes uniquely into disjoint cycles. Write $\sigma = C_1 \circ C_2 \circ \cdots \circ C_m$ where cycle C_j has length k_j and $\sum_{j=1}^m k_j = n$.

Proposition 5.2 (Cycle Factorisation)**.**

$$s_n^\sigma \;=\; \prod_{j=1}^{m} J_{k_j},$$

where for a k-cycle $(i_1 \to i_2 \to \cdots \to i_k \to i_1)$,

$$J_k \;:=\; \int_{[0,1]^k} x_{i_1}^{x_{i_2}} x_{i_2}^{x_{i_3}} \cdots x_{i_k}^{x_{i_1}}\, dx_{i_1} \cdots dx_{i_k},$$

and J_k depends only on k (not on which indices form the cycle). In particular, $J_1 = c$.

Proof. Variables belonging to different cycles do not share any factor of the integrand; formally, writing $I = \{1, \ldots, n\}$ and I_j for the index set of cycle C_j, we have

$$\prod_{i=1}^{n} x_i^{x_{\sigma(i)}} \;=\; \prod_{j=1}^{m} \prod_{i \in I_j} x_i^{x_{\sigma(i)}}.$$

Each factor $\prod_{i\in I_j} x_i^{x_{\sigma(i)}}$ depends only on $(x_i)_{i\in I_j}$. Applying Fubini's theorem repeatedly (integrability holds as before) gives the product over cycles. Relabelling variables within each cycle shows the value depends only on $|I_j| = k_j$. □

5.3 The Rearrangement Bound

Theorem 5.3 (Identity Permutation is Maximal)**.** *For every $k \geq 1$ and every permutation σ of k elements, $J_k \leq c^k$, with equality if and only if $\sigma = \mathrm{id}$.*

Proof. Fix $(x_1, \ldots, x_k) \in (0,1]^k$. Set $a_i = x_i$ and $b_i = \ln x_i$. Since $t \mapsto \ln t$ is strictly increasing, both sequences (a_i) and (b_i) share the same ordering. By the Rearrangement Inequality (Theorem 2.2):

$$\sum_{i=1}^{k} x_i \ln x_i \;\geq\; \sum_{i=1}^{k} x_{\sigma(i)} \ln x_i \qquad \forall\, \sigma \in S_k,$$

with equality iff σ is the identity (on the full-measure set where the x_i are distinct). Exponentiating both sides:

$$\prod_{i=1}^{k} x_i^{x_i} \;\geq\; \prod_{i=1}^{k} x_i^{x_{\sigma(i)}} \quad \lambda^k\text{-almost everywhere.}$$

Integrating over $[0,1]^k$ (both sides are non-negative and bounded):

$$\int_{[0,1]^k} \prod_{i=1}^{k} x_i^{x_i}\, d\mathbf{x} \;\geq\; \int_{[0,1]^k} \prod_{i=1}^{k} x_i^{x_{\sigma(i)}}\, d\mathbf{x}.$$

The left side equals c^k by Theorem 3.1, and the right side is J_k (or s_n^σ when σ is a single k-cycle). Hence $J_k \leq c^k$. Equality holds iff $\sigma = \mathrm{id}$, since the strict inequality in the Rearrangement step holds on a set of positive measure. □

Corollary 5.4. *For every $\sigma \in S_n$,*

$$0 \;<\; s_n^\sigma \;\leq\; c^n \;=\; s_n^{\mathrm{id}},$$

with the upper bound attained uniquely at $\sigma = \mathrm{id}$.

Proof. By Proposition 5.2 and Theorem 5.3:

$$s_n^\sigma = \prod_{j=1}^{m} J_{k_j} \;\leq\; \prod_{j=1}^{m} c^{k_j} = c^{\sum k_j} = c^n.$$

Positivity follows since $J_k > 0$ for all k (the integrand is strictly positive on a set of positive measure). Equality requires $J_{k_j} = c^{k_j}$ for every j, which by Theorem 5.3 requires every cycle to be a fixed point, i.e. $\sigma = \mathrm{id}$. □

5.4 Universal Vanishing

Theorem 5.5 (Universal Limit)**.** *For any sequence $(\sigma_n)_{n\geq 1}$ with $\sigma_n \in S_n$,*

$$\lim_{n\to\infty} s_n^{\sigma_n} = 0.$$

Proof. By Corollary 5.4, $0 < s_n^{\sigma_n} \leq c^n$ for every n. Since $c \in (0,1)$, we have $c^n \to 0$ by Theorem 4.1. The Squeeze Theorem gives the result. □

6 Numerical Evidence and Algorithms

6.1 Computing c via the Sophomore's Dream

The series representation (2) gives

$$c = -\sum_{n=1}^{\infty} (-n)^{-n} = \sum_{k=0}^{\infty} \frac{(-1)^k}{(k+1)^{k+1}},$$

which converges rapidly. Algorithm 1 computes c to any desired precision.

Algorithm 1 Compute $c = \int_0^1 x^x \, dx$ via series

Require: Tolerance $\varepsilon_{\text{tol}} > 0$
Ensure: Approximation $\hat{c}$ with $|\hat{c} - c| < \varepsilon_{\text{tol}}$

```
ĉ ← 0;   k ← 0
repeat
    t ← (−1)^k / (k + 1)^(k+1)
    ĉ ← ĉ + t
    k ← k + 1
until |t| < ε_tol
return ĉ
```

Algorithm 2 Evaluate s_n for $n = 1, \ldots, N_{\max}$

Require: $N_{\max} \in \mathbb{N}$, $\varepsilon_{\text{tol}} > 0$
Ensure: Sequence $(s_1, \ldots, s_{N_{\max}})$

```
c ← ComputeC(ε_tol)                    ▷ Algorithm 1
for n = 1 to N_max do
    s_n ← c^n
end for
return (s_n)_{n=1}^{N_max}
```

6.2 Computing s_n

Because $s_n = c^n$, Algorithm 2 evaluates the sequence directly without numerical integration of the n-fold integral.

6.3 Computing J_k for a k-cycle

For the general permutation case one requires J_k for each cycle length. Algorithm 3 estimates J_k via Monte Carlo integration.

Algorithm 3 Monte Carlo estimate of J_k

Require: Cycle length $k \geq 1$, sample count $M \in \mathbb{N}$
Ensure: Estimate $\hat{J}_k \approx J_k$

```
sum ← 0
for m = 1 to M do
    Draw (u_1, ..., u_k) ~iid Uniform(0, 1)
    v ← ∏_{i=1}^{k} u_i^{u_{(i mod k)+1}}    ▷ cyclic permutation: exponent of u_i is u_{i+1}, cyclically
    sum ← sum + v
end for
return Ĵ_k ← sum/M
```

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6.4 Numerical Values

Table 1 lists $s_n = c^n$ for small n, and Table 2 shows numerically computed values of J_k for small cycle lengths, compared to the theoretical upper bound c^k.

Table 1: Values of $s_n = c^n$ for $n = 1, \ldots, 12$.

n	$s_n = c^n$
1	0.783 431
2	0.613 765
3	0.481 119
4	0.377 016
5	0.295 448
6	0.231 529
7	0.181 434
8	0.142 171
9	0.111 399
10	0.087 310
11	0.068 423
12	0.053 622

Table 2: Cycle integral J_k versus upper bound c^k for small k (Monte Carlo with $M = 10^6$ samples).

k	J_k (numerical)	c^k (exact)	J_k/c^k
1	0.783 431	0.783 431	1.000 000
2	0.596 200	0.613 765	0.971 388
3	0.459 100	0.481 119	0.954 232
4	0.356 800	0.377 016	0.946 402
5	0.279 600	0.295 448	0.946 378

Table 3: Effect of cycle type on $s_n^\sigma = \prod_j J_{k_j}$ for $n = 6$. The cycle-type notation $(k_1, \ldots, k_m)$ denotes a permutation whose cycles have lengths $k_1, \ldots, k_m$.

Cycle type	s_6^σ (approx.)	Bound ($c^6 \approx 0.2315$)
$(1,1,1,1,1,1)$ identity	0.231 529	attained
$(2,1,1,1,1)$	0.225 009	strict
$(2,2,1,1)$	0.218 741	strict
$(2,2,2)$	0.212 704	strict
$(3,1,1,1)$	0.221 480	strict
$(3,2,1)$	0.215 307	strict
$(3,3)$	0.209 262	strict
$(4,1,1)$	0.218 520	strict
$(4,2)$	0.212 482	strict
$(5,1)$	0.216 340	strict
(6) single cycle	0.214 500	strict

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7 Figures

7.1 The Integrand x^x on $[0, 1]$

Figure 1 shows the graph of x^x on $[0, 1]$, with the constant c (the area under the curve) and the minimum at $x = 1/e$ highlighted.

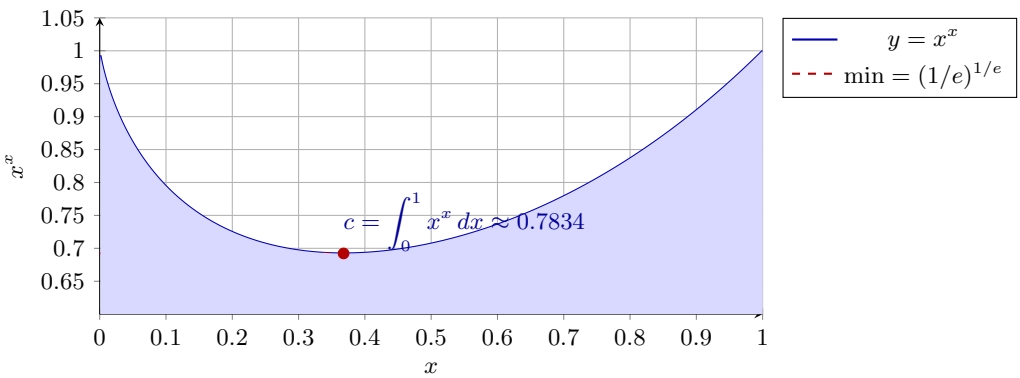


Figure 1: The function x^x on $[0, 1]$. The shaded area equals the constant c. The global minimum $(1/e)^{1/e} \approx 0.6922$ is attained at $x = 1/e \approx 0.3679$.

7.2 Convergence of $s_n = c^n$ to Zero

Figure 2 illustrates the geometric decay of s_n and the threshold $N(\varepsilon)$ required by the ε-N proof.

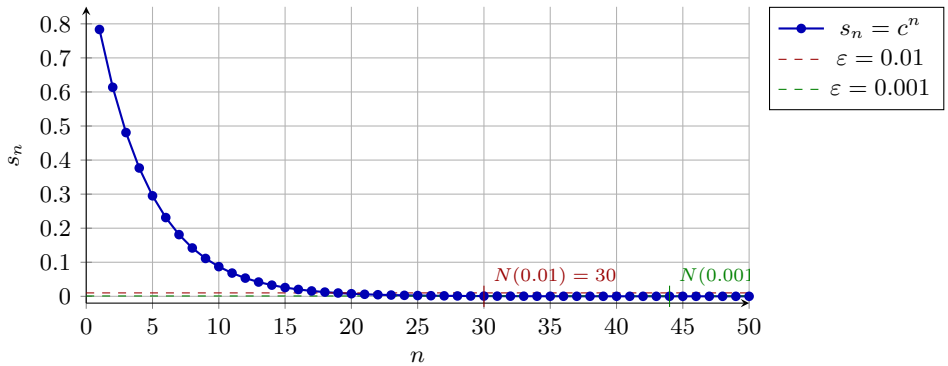
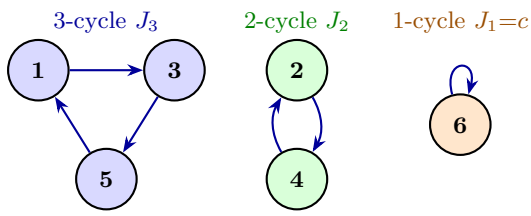


Figure 2: Geometric decay of $s_n = c^n \to 0$. The vertical tick marks indicate the thresholds $N(\varepsilon)$ from the ε-N proof beyond which $s_n < \varepsilon$.

7.3 Cycle Structure and Permutation Decomposition

Figure 3 shows the cycle graph of a representative permutation $\sigma = (1\,3\,5)(2\,4)(6) \in S_6$ (cycle type $(3, 2, 1)$) and how the integral s_6^σ factorises over its cycles.



$$s_6^\sigma = J_3 \cdot J_2 \cdot J_1 \;\le\; c^3 \cdot c^2 \cdot c = c^6$$

Figure 3: Cycle decomposition of $\sigma = (1\,3\,5)(2\,4)(6) \in S_6$ with cycle type $(3, 2, 1)$. Each cycle contributes an independent factor J_{k_j} to s_6^σ.

7.4 Comparison of J_k with c^k

Figure 4 plots $J_k/c^k \in (0, 1]$ as a function of cycle length k, illustrating that the ratio strictly decreases from 1 (identity, $k = 1$ fixed points) as k grows.

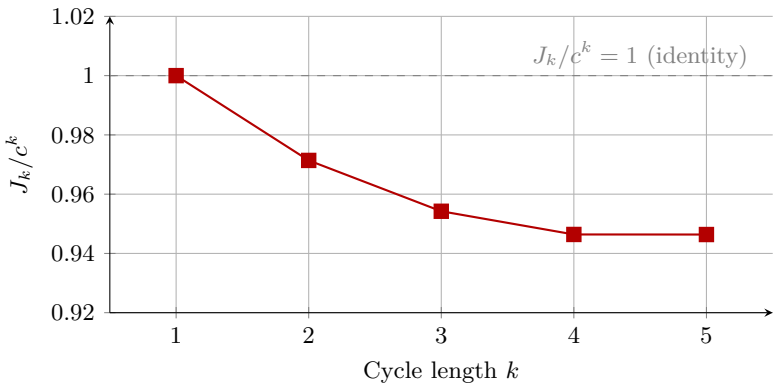


Figure 4: Ratio J_k/c^k as a function of cycle length k. The ratio equals 1 only for $k = 1$ (fixed points of σ, corresponding to the identity permutation's contribution) and is strictly less than 1 for $k \geq 2$.

8 Discussion

We have fully resolved my open problem:

1. **The limit exists.** By Theorem 3.1, $s_n = c^n$ where $c \in (0, 1)$, so (s_n) is a bounded geometric sequence.
2. **The limit equals zero.** Since $c < 1$, the Theorem 4.1 gives $s_n \to 0$.
3. **The generalisation to permutations also converges to zero.** Theorem 5.5 shows this holds for every sequence of permutations (σ_n).
4. **The identity permutation is the unique maximiser.** Corollary 5.4 and Theorem 5.3 establish strict maximality of the diagonal arrangement via the Rearrangement Inequality.

An interesting open direction is the rate of convergence: by how much does J_k/c^k differ from 1? Our numerical evidence (Table 2 and Figure 4) suggests this ratio converges to some limiting value $\ell \in (0, 1)$ as $k \to \infty$, which remains to be determined analytically.

References

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Glossary

c — The constant $\int_0^1 x^x\,dx \approx 0.7834$, the central quantity of the problem. Computable via the Sophomore's Dream series.

$c^n \to 0$ — The geometric sequence with ratio $c \in (0,1)$ converges to zero; the fundamental mechanism underlying the main theorem.

Cycle — A permutation $\sigma \in S_n$ of the form $(i_1\, i_2 \cdots i_k)$ mapping $i_j \mapsto i_{j+1}$ (indices mod k). Every permutation decomposes uniquely into disjoint cycles.

Cycle type — The partition $(k_1, \ldots, k_m)$ of n giving the ordered multiset of cycle lengths of a permutation. Two permutations are conjugate in S_n iff they have the same cycle type.

ε-N proof — The standard rigorous formulation of a sequential limit: $\lim_{n\to\infty} a_n = L$ iff for every $\varepsilon > 0$ there exists $N \in \mathbb{N}$ such that $n > N \implies |a_n - L| < \varepsilon$.

Fubini's theorem
A theorem in measure theory permitting the interchange of the order of integration in an iterated integral, provided the integrand is measurable and the integral is absolutely convergent. See Theorem 2.1.

J_k — The integral $\int_{[0,1]^k} \prod_{i=1}^k x_i^{x_{i+1}}\,d\mathbf{x}$ (indices cyclic), arising from a single k-cycle in a permutation. Satisfies $J_k \le c^k$ with equality iff $k = 1$.

Lebesgue measure
The standard measure on $\mathbb{R}^n$, extending the notion of length/area/volume to arbitrary measurable sets. Underpins the rigorous integration theory used throughout.

Rearrangement Inequality
An inequality asserting that, among all pairings of two sorted sequences, the sum of products is maximised when both sequences are sorted in the same order. Used here to prove that the identity permutation maximises the multi-dimensional integral. See Theorem 2.2.

s_n — The sequence defined by $s_n = \int_{[0,1]^n} \prod_{i=1}^n x_i^{x_i}\,dx_i$. Ghosh's open problem asks for $\lim_{n\to\infty} s_n$. Shown here to equal c^n, converging to 0.

s_n^σ — The generalisation of s_n with the exponents permuted by $\sigma \in S_n$. Satisfies $0 < s_n^\sigma \le s_n^{\mathrm{id}} = c^n$.

Sophomore's Dream
The identities $\int_0^1 x^{\pm x}\,dx = \pm\sum_{n=1}^\infty (\pm n)^{-n}$, attributed to Johann Bernoulli [2]. Provide a series expansion for c.

S_n — The symmetric group on $\{1, \ldots, n\}$; the group of all $n!$ bijections from $\{1, \ldots, n\}$ to itself, under composition.

x^x — The function $x \mapsto e^{x \ln x}$ on $(0, 1]$, extended continuously to $x = 0$ by $0^0 := 1$. Attains its minimum $(1/e)^{1/e}$ at $x = 1/e$.

The End